Near-Field Beamforming and Doppler Compensation for mmWave Internet of Vehicles

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***Abstract*— This letter considers the near-field beamforming in millimeter wave Internet-of-vehicle, where phased arrays have been widely used for mitigating misfocus effect. In near-field regime, the angle and distance need to be considered simultaneously due to the fact that the wave propagation is spherical. We design the phase profile of the antenna arrays by using the angle and distance information to meet characteristic of near-field spherical wavefront. Then, we construct the chirp signal based on the hybrid frequency modulation technique to improve Doppler sensitivity to compensate the effect caused by the movement of the vehicles. Our results reveal that the proposed method can achieve high data rates by reducing the beam misfocus and Doppler effect in millimeter wave Internet-of-vehicle.**

***Index Terms*— Multiple-input multiple-output, beamforming, near-field communication, mmWave, Doppler effect.**

I. Introduction

MILLIMETER wave (mmWave) and extremely large-scale multiple-input multiple-output (XL-MIMO) are considered to be promising technologies [1] to meet the key requirements in 6G, including ultra-high throughput, ultra-low latency, and ultra-high reliability, etc. In the mmWave systems, antenna arrays always are used at both transmitter and receiver. This is the case where using XL-MIMO is highly desirable. XL-MIMO uses hundreds or thousands of antennas at the base station, i.e., increasing the antenna number/size, resulting in a large near-field range on the order of hundreds of meters that can not be neglected [2], which is of crucial significance for the next-generation communication [3].

There are many challenges in the near-field regime. The reason is that the propagation properties of the far-field and near-field is different. The former can be reliably approximated as planar and the latter propagate exhibiting a non-negligible spherical wavefront [4]. Under the spherical wave assumption, many issues need to be emphatically considered and intelligently designed, such as channel model, channel estimation [5], [6], and beamforming, etc. Beamforming is an essential technology in mmWave Internet-of-vehicle to achieve a higher data rate. The poor beamformers lead to the misfocus effect and further reduce the data rate in the near-field. Myers and Heath [7] point out the misfocus effect causing an obvious rate loss. They present a beamforming method by using the angle and distance information for improving the performance of transmission rate of data. For multiuser MIMO communication, the analysis of different antenna structures are studied in [8] within near-field regime, like fully-digital architectures and the emerging dynamic metasurface antenna. The influence of user's location, the carrier frequency and the array dimension are determined in [9] for beamforming in Terahertz linear antenna array. Considering the passive nature in respect of signal reflection, the large-scale antenna arrays is required in Intelligent reconfigurable surfaces, which brings challenges about near-field effect [10].

In the existing near-field beamforming, it is assumed that locations of the receive antennas are fixed or with a low speed. That is, the aforementioned studies have not considered the movement that induces Doppler shift. However, the mobility and the Doppler shift are important in mmWave Internet-of-vehicle scenarios [11]. Considering the propagation properties of the near-field, we design the beamformer on the basis of the InFocus scheme [7]. InFocus is a low-complexity spatial coding technique that does not require any iteration, which is proposed for mitigating misfocus in near-field communication. But, it uses the non-linear frequency modulated (NLFM) continuous waveform chirp to construct the signal, which leads to poor Doppler tolerance [12] and has been limited to the considered high-mobility communication scenarios [13].

To fill in this gap, we propose a Doppler compensation beamforming scheme (DCBS) for vehicles moving in the near-field regime for mmWave Internet-of-vehicle. In this letter, we investigate the problem of transmit beamforming with a uniform circular planar phased array. The effects of spherical wavefronts and beam misfocus in the near-field are considered, and we analyze the transmit beamforming vector design leveraging the spatial coding approach [7]. Both angle and location information are incorporated to mitigate beam misfocus effects. We then design the hybrid non-linear frequency modulation (hybrid-FM) beamforming profile to balance between windowed LFM and unwindowed NLFM, in order to address the non-linear change of the amplitude modulation effect and the Doppler shift. The performance of the proposed DCBS is evaluated in terms of the achievable rate. Numerical results validate that the proposed method

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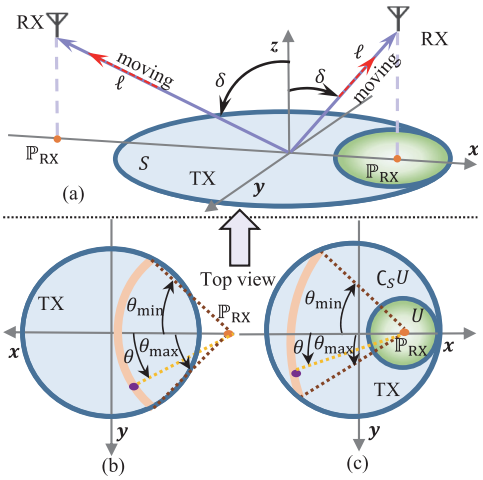

Fig. 1. A illustration of the system in the near-field: (a) the RX and TX array in the mathematical coordinate space, where S denotes the set of the antenna elements in the x-y plane and its center is $(0, 0, 0)$, and the RX lies in $(-\ell \sin\delta, 0, \ell\cos\delta)$, and the projection $\mathbb{P}_{\text{RX}}$ can or not in the S zone and its location is $(-\ell\sin\delta, 0, 0)$; (b) the top view of the system that $\mathbb{P}_{\text{RX}}$ is out of the S zone; (c) the top view of the system that $\mathbb{P}_{\text{RX}}$ lies in the S zone.

enables phased arrays to achieve higher rates than comparable techniques at different vehicle moving speeds.

The rest of this letter is organized as follows. The channel model for mmWave Internet-of-vehicle in near-field regime is described in Section II. Then, we propose our method DCBS based on InFocus scheme and hybrid-FM in Section III. Finally, simulation results and conclusions are shown in Section IV and Section V.

II. System Model

In this letter, we focus on the problem of the transmit beamforming in a multiple input single output (MISO) system for the line-of-sight (LoS) scenario. We assume that the accurate channel state information and the initial localization information have been acquired by the related works [5], [6]. We consider the transmitter (TX) is a circular planar antenna array [14] with a radius of R to focus radio frequency (RF) signals, which consists of N_t isotropic antennas placed in the x-y plane of a Cartesian coordinate system. The geometric center of the TX array is aligned with the origin of the coordinate system, which is $(0, 0, 0)$. The TX equips a single RF chain connected N_t phase shifters. We maximize the signal power of the receiver with the operation of the phase shifters in the phased array.

As shown in Fig. 1(a), ℓ denotes the distance from the origin to the receiver antenna (RX), and δ represents the angle excursion of the ray joining the center of the TX array and the RX from the boresight direction to the target position. Due to the "symmetric" nature of the circular transmit array, without loss of generality, we assume that the RX lies in the x-z plane, i.e., the location is $(-\ell\sin\delta, 0, \ell\cos\delta)$. We use $\mathcal{L}(x, y, \ell, \delta)$ to denote the distance between the antenna element located in (x, y) of TX and RX as shown in Fig. 1, which is obtained as

$$\mathcal{L}(x, y, \ell, \delta) = \sqrt{(x + \ell\sin\delta)^2 + y^2 + \ell^2\cos^2\delta}. \tag{1}$$

The channel between the antenna located in (x, y) and the RX antenna for a frequency of f can be expressed as [15]

$$h(f, x, y) = \frac{c}{2\pi f\mathcal{L}(x, y, \ell, \delta)} \exp\left\{-j\frac{2\pi f\mathcal{L}(x, y, \ell, \delta)}{c}\right\}, \tag{2}$$

where, $j = \sqrt{-1}$, and c is the speed of light.

The mobility of the receiver leads to the poor beamforming gain and lower data transmission rates [16], [17]. Considering this, (2) can be rewritten as

$$h(t, f, x, y) = \frac{\exp\left\{j2\pi\left(\frac{vt\cos(\eta)}{\lambda_c} - \frac{f\mathcal{L}(x,y,\ell(t),\delta)}{c}\right)\right\}}{2\pi f c^{-1}\mathcal{L}(x, y, \ell(t), \delta)}, \tag{3}$$

where, t is the time variable, v is the vehicle velocity, λ_c denotes the wavelength, η denotes the angle between direction of motion, and $\ell(t)$ equals $\ell + vt$. It is observed that $2\pi vt\cos(\eta)/\lambda_c \in [-\pi, \pi]$.

Let us assume that the RX moves straight keeping the angle δ invariable. We use $\nu = 2\pi vt\cos(\eta)/\lambda_c$ to denote the phase shift and $\ell' = \ell(t)$ to denote the distance after moving, and then (3) is simplified to

$$h(f, x, y) = \frac{c}{2\pi f\mathcal{L}(x, y, \ell')} \exp\left\{j\left(\nu - \frac{2\pi f\mathcal{L}(x, y, \ell')}{c}\right)\right\}. \tag{4}$$

In a short range LoS system, the TX controls RF signals for transmission through the antennas in the phased array architecture by applying beamforming weights. For an antenna, we assume that the phase of the RF signals is the only thing that can be changed by the TX. Let $b(x, y)$ denote the beamforming weight applied by the transmitter antenna located at (x, y) coordinates for directing the signals towards the RX, as [18]

$$b(x, y) = \frac{1}{\sqrt{\pi R^2}} \exp(j\phi(x, y)), \tag{5}$$

in which, $\phi(x, y)$ is the phase shift of the element of the TX.

III. Proposed Method

In this section, we introduce a spatial coding scheme for robust transmit beam alignment under misfocus conditions. Considering the movement of the RX, an approach based on hybrid-FM is designed for achieving higher data rates in the near-field region.

The beamforming gain is low with the standard beamforming method at different frequencies, resulting in receiving only small bandwidth signals that are not suitable for achieving high data rates. It stands to the reason that the misfocus effect in phased arrays. Thus, the spatial coding approach named InFocus [7] is proposed with low complexity.

All RF signals will be summed in the RX. We use $J(f)$ to denote the equivalent frequency domain channel at the frequency f between the TX and the RX. Considering an imaginary transmitter (ITX) [7], [19] equipped with a continuous aperture of radius R, i.e., the number of antenna elements is infinite in the array. $J(f)$ then can be obtained as

$$J(f) = \int_S h(f, x, y)b(x, y)\mathrm{d}x\mathrm{d}y, \tag{6}$$

where, $S = \{(x, y) : x^2 + y^2 \leq R^2\}$ denotes area including the antenna elements that fulfill the condition $x^2 + y^2 \leq R^2$ and $z = 0$ in the TX array, as shown in Fig. 1.

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Substituting (1), (3) and (5), the integral in (6) is

$$J(f) = c\int_S \frac{\exp\{j(\phi(x,y)+\nu-\frac{2\pi f\mathcal{L}(x,y,\ell')}{c})\}}{2\ fR\Delta\mathcal{L}(x,y,\ell')\pi^{3/2}}\mathrm{d}x\mathrm{d}y, \quad (7)$$

where Δ is the spacing between continuous antenna elements along the x and y dimensions. Intuitively, $J(f)$ is the equivalent channel that depends on the phase $\phi(x,y)$.

Standard beamforming $\varphi(x,y) = 2\pi f_c\mathcal{L}(x,y,\ell')/c$ is a solution that can maximize the energy $|J(f_c)|^2$ at the center frequency f_c. However, it follows that when the arrays response varies substantially with the frequency in the large antenna array, there would be much misfocus in phased arrays. The reason is that $\varphi(x,y)$ changes with f_c. For mitigating misfocus and improving the data rate, we need a phase profile $\phi(x,y)$ that maximize the energy of the equivalent channel response in the entire frequency band. Thus, we define $\zeta(x,y)$ as a phase profile, which is designed and added to standard beamforming solution $\varphi(x,y)$ for comprising the whole phase profile $\phi(x,y)$. In (7), we substitute $\phi(x,y) = \varphi(x,y) + \zeta(x,y)$ and $\varphi(x,y) = 2\pi f_c\mathcal{L}(x,y,\ell')/c$ to write

$$J(f)=\int_S \frac{\exp\{j(\zeta(x,y)+\nu-2\pi(f-f_c)\frac{\mathcal{L}(x,y,\ell')}{c})\}}{2\ fR\Delta\pi^{3/2}c^{-1}\mathcal{L}(x,y,\ell')}\mathrm{d}x\mathrm{d}y \quad (8)$$

$$\overset{(a)}{=} c\int_S \frac{\exp\{j(\zeta(x,y)+\nu-\omega\mathcal{L}(x,y,\ell'))\}}{2fR\Delta\mathcal{L}(x,y,\ell')\pi^{3/2}}\mathrm{d}x\mathrm{d}y, \quad (9)$$

where (a) holds due to the defination that the spatial frequency is $\omega = 2\pi(f-f_c)/c$.

Obviously, $c(\sqrt{\pi}R\Delta)^{-1}$ is constant that has no effect on the design of the phase profile, therefore, we ignore it to simplify (9) as

$$\widetilde{J}(f) = \frac{1}{2\pi f}\int_S \frac{\exp\{j(\zeta(x,y)+\nu-\omega\mathcal{L}(x,y,\ell'))\}}{\mathcal{L}(x,y,\ell')}\mathrm{d}x\mathrm{d}y, \quad (10)$$

$$\overset{(a)}{=} \int_S \frac{\exp\{j(\zeta(x,y)+\nu-\omega\mathcal{L}(x,y,\ell'))\}}{2\pi\mathcal{L}(x,y,\ell')}\mathrm{d}x\mathrm{d}y, \quad (11)$$

where (a) holds due to the fact that the f^{-1} scaling results in a smaller variation in beamforming gain compared to the integral in (10). For example, considering a system over a bandwidth of $B = 40$ GHz around a carrier frequency of $f_c = 300$ GHz, the variation in $|J(f)|^2$ is $20\log_{10}(320/280) \approx 1.2$ dB due to the f^{-1} term [7]. The subsequent analysis and design of the phase profile are on the basis of this scaled version of the equivalent frequency domain channel.

In the first case, the RX is not along the boresight direction and its projection $\mathbb{P}_{\text{RX}}$ is out of the S zone on x-y plane as shown in Fig. 1(b). We represent the ITX array in a polar coordinate plane and substitute $\rho = \sqrt{(x+\ell'\sin\delta)^2+y^2}$ and $\mathcal{L}(x,y,\ell') = \sqrt{\rho^2+\ell'^2\cos^2\delta}$ in (11). Now, $\mathbb{P}_{\text{RX}}$ is the center of the polar coordinate system. Let θ be the angle between the x-axis and the straight line which is between the $\mathbb{P}_{\text{RX}}$ and the ith antenna element, as seen in Fig. 1(b). Then, we have

$$\widetilde{J}(f) = \int_{\rho_l}^{\rho_u}\int_{\theta_l}^{\theta_u} \frac{e^{j(\zeta(x,y)+\nu-\omega\sqrt{\rho^2+\ell'^2\cos^2\delta})}}{2\pi\sqrt{\rho^2+(\ell'\cos\delta)^2}}\rho\mathrm{d}\rho\mathrm{d}\theta \quad (12)$$

$$\overset{(a)}{=} \int_{q_l}^{q_u} \frac{(\theta_u-\theta_l)\exp\{j(\zeta(q)+\nu-\omega q)\}}{2\pi}\mathrm{d}q \quad (13)$$

$$\overset{(b)}{=} \int_{q_l}^{q_u} \Upsilon(q)\exp\{j(\zeta(q)-\omega q)\}\mathrm{d}q, \quad (14)$$

where, l and u denote the lower and upper bound, $\rho \in [\ell'\sin\delta - R, \ell'\sin\delta + R]$ and $\theta(\rho) = \arcsin(y/\rho)$, and (a) follows from the definition that $q = \sqrt{\rho^2+\ell'^2\cos^2\delta}$, and (b) denotes a operation substituting $\Upsilon(q) = e^{j\nu}(\theta_u-\theta_l)/2\pi$. For simplicity of notation, we apply $\omega = 2\pi(f-f_c)/c$ and define

$$\mathcal{J}(\omega) = \int_{q_l}^{q_u} \Upsilon(q)\exp\{j(\zeta(q)-\omega q)\}\mathrm{d}q. \quad (15)$$

For a wideband mmWave system, the phase profile is designed to achieve an approximately flat $|\mathcal{J}(\omega)|^2$ over the desired frequency range $[-\pi B/c, \pi B/c]$. From (15), it's clear that $|\mathcal{J}(\omega)|^2$ is not linear due to the $\Upsilon^2(q)$, that is, the beamforming gain is induced by the amplitude modulation effect [7]. Thus, we use the hybrid-FM chirp along the spatial dimension to construct the signal. The hybrid-FM is a solution to the Doppler sensitivity problem, which uses amplitude windowing and NLFM. It represents a compromise between windowed LFM and unwindowed NLFM.

The principle of stationary phase is adopted to have an asymptotic approximation of the integral with the fact that $\Upsilon(q)$ varies slowly contrasted with phase [12], [20], [21]. We expand $\zeta(q)$ in a 3-term Taylor's series near the point of stationary phase q, where $\zeta'(q) = 0$. Substituting it in (15), we have

$$|\mathcal{J}(\omega_q)|^2 \simeq 2\pi\Upsilon^2(q)|\zeta''(q)|^{-1}, \quad (16)$$

in which, $\omega_q = \zeta'(q)$ is the instantaneous frequency at q, and $\zeta''(q)$ is the second derivative of $\zeta(q)$ with respect to q.

Considering the Doppler shift caused by the moving vehicles, and the implementation with fairly simple calculations, we select the most promising solution that combines LFM and tan-FM [12], [22], [23], which is expressed as

$$\zeta'(q) = \pi B\left[\frac{2(1-\gamma_1)q}{Q} + \frac{\gamma_1\tan(2q\gamma_2/Q)}{\tan(\gamma_2)}\right], \quad (17)$$

where γ_1 is the ratio of nonlinear modulation, and γ_2 controls the proportion of the tangential curve, and $Q = q_u - q_l$. For misfocus robust beamforming, it has to concentrate the spectrum of the chirp in $[-\pi B/c, \pi B/c]$, i.e., demanding $\omega_{q_l} = -\pi B/c$ and $\omega_{q_u} = \pi B/c$. Then, the values of γ_1 and γ_2 can be obtained.

Next, differentiating (17) to find $\zeta''(q)$ as follows:

$$\zeta''(q) = \frac{2\pi B}{Q}\left[1-\gamma_1+\frac{\gamma_1\gamma_2(1+\tan^2(2q\gamma_2/Q))}{\tan\gamma_2}\right]. \quad (18)$$

The Hann window is chosen as a desired shape of the power spectrum of the signal, which is given by:

$$|\mathcal{J}(\omega_q)|^2 = \frac{1}{2}\left[1-\cos(2\pi q/Q)\right]. \quad (19)$$

Thus, the amplitude modulation function $\Upsilon(q)$ as follows

$$\Upsilon(q) \simeq \left\{\frac{B}{2Q}\left[1-\cos(2\pi q/Q)\right] \times \left[1-\gamma_1+\frac{\gamma_1\gamma_2(1+\tan^2(2q\gamma_2/Q))}{\tan\gamma_2}\right]\right\}^{1/2}. \quad (20)$$

The designed phase profile $\zeta(q)$ is the solution to the differential equation $\zeta'(q)$ in (17) that can be acquired by using numerical integration, i.e., $\zeta(q) = \int_{q_l}^{q}\zeta'(q)\mathrm{d}q$. Then,

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the designed 2D phase profile $\zeta(x, y)$ is obtained easily by combining $\zeta(q)$ and $q = \sqrt{(x + \ell' \sin\delta)^2 + y^2 + \ell'^2 \cos^2\delta}$.

In the second case, the projection $\mathbb{P}_{\text{RX}}$ is in the S area as shown in Fig. 1(c). The value of $\zeta(x, y)$ is same for the antenna elements that are the same distance from $\mathbb{P}_{\text{RX}}$. According to the value of δ, we design the phase profile in two situations.

1) Along the Boresight: The beamforming is directed along the boresight of the arrays, i.e., $\delta = 0°$. Now, we discuss the design process of constructing the signal with the LFM chirp.

When $\delta = 0°$, (1) is rewritten as $\sqrt{x^2 + y^2 + \ell'^2}$ with the definition that $x = r\cos\beta$ and $y = r\sin\beta$ are in the polar coordinate representation. We then have $\zeta(r, \beta)$ where $r = \sqrt{x^2 + y^2}$ and $\beta = \arctan(y/x)$. After deeper consideration, we realize the designed phase profile only depends on the distance from $\mathbb{P}_{\text{RX}}$, thus $\zeta(r, \beta) = \zeta(r)$. Due to the fact that the phase profile is designed irrespective of the angle $\beta \in [0, 2\pi]$ with the definition $q = \sqrt{r^2 + \ell'^2}$, we rewrite (11) as

$$\widetilde{J}(f) = \int_{\ell'}^{\sqrt{\ell'^2 + R^2}} \exp j(\zeta(q) + \nu - \omega q)\mathrm{d}q, \tag{21}$$

where $r \in [0, R]$. Similar to the operation in (15), we have

$$\mathcal{J}(\omega) = \int_{\ell'}^{\sqrt{\ell'^2 + R^2}} \exp j(\zeta(q) + \nu - \omega q)\mathrm{d}q. \tag{22}$$

For a linear FMCW chirp, the instantaneous frequency of the signal increases linearly in the frequency range, here is $[q_l, q_u] = [\ell', \sqrt{\ell'^2 + R^2}]$. Thus, we design the $\zeta(q)$ to make its derivative increases linearly, that is, $\zeta'(q_l) = -\pi B/c$, and $\zeta'(q_u) = \pi B/c$. According to the form $\zeta(q) = \sigma_1 q + \sigma_2 q^2$, where σ_1 and σ_2 are the constant parameters, and we have $\sigma_1 + 2\sigma_2 q_l = -\pi B/c$, $\sigma_1 + 2\sigma_2 q_u = \pi B/c$. After finding the solution of this, the phase profile is given as

$$\zeta(q) = -\frac{\pi B}{c}\left[\frac{\sqrt{\ell'^2 + R^2} + \ell'}{\sqrt{\ell'^2 + R^2} - \ell'} q - \frac{q^2}{\sqrt{\ell'^2 + R^2} - \ell'}\right]. \tag{23}$$

We can calculate the 2D phase profile $\zeta(x, y)$ by substituting $q = \sqrt{x^2 + y^2 + \ell'^2}$ into (23).

2) Not Along the Boresight: We divide the S area into two parts—U and $\complement_S U$. The U denotes the zone where elements are within a distance of $(R - \ell' \sin\delta)$ from $\mathbb{P}_{\text{RX}}$, and then the rest of S area belong to $\complement_S U$ area.

For U, it is obvious that the beamforming is in a boresight scenario and the phase profiles can be obtained by using (23). For $\complement_S U$, the phase profile can be computed with the analysis of the first case. In consequence, the phase profile of the whole S area, which is given in the following form

$$\mathcal{J}(\omega) = \int_{\ell' \cos\delta}^{\sqrt{(\ell' \sin\delta + R)^2 + (\ell' \cos\delta)^2}} \Gamma(q) e^{j(\zeta(q) - \omega q)}\mathrm{d}q, \tag{24}$$

in which,

$$\Gamma(q) = \begin{cases} \Upsilon(q), & \ell' \cos\delta \le q \le q_l \\ e^{j\nu}, & q_l < q \le q_u \end{cases}, \tag{25}$$

where, $q_l = \sqrt{(\ell' \sin\delta - R)^2 + (\ell' \cos\delta)^2}$, and $q_u = \sqrt{(\ell' \sin\delta + R)^2 + (\ell' \cos\delta)^2}$. Next, (24) can be solved using the hybrid design method in the first case as mentioned before.

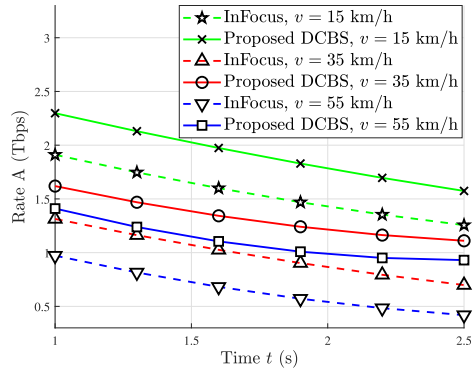

Fig. 2. Achievable rate with time t at different vehicle velocity v when the angle $\delta = 0$. The transceiver distance grows large over time.

IV. Simulation Results

In the simulations, we assume the location information of the RX is known to the antenna elements at the TX. We use the carrier frequency of $f_c = 300$ GHz, the bandwidth $B = 40$ GHz, and the speed of light $c = 3 \times 10^8$ m/s. At the TX, we set the spacing between the antenna elements to $\Delta = 0.5$ mm, the number of antennas is $N_t = 124980$, and the radius of the circular planar array is 10 cm. We set the TX array's total transmit power to 1 mW, and the transmit power of each antenna is $1/N_t$ mW.

We compare our proposed DCBS with the near-field beamforming approach InFocus [7], which is a spatial coding approach based on the principle of stationary phase with low complexity for mitigating misfocus effect in phased arrays.

According to [24], the concept is described here to compute the achievable rate of an equivalent channel. The wideband channel is separated into N_a parts. We use $P_a = \sum_{i=1}^{N_a} P_i$ to denote the total transmit power, and P_i is the transmit power allocated over the ith sub-band. Under this condition, the achievable rate of an equivalent channel is given by

$$\mathrm{A} = \frac{B}{N_a}\sum_{i=1}^{N_a} \log_2\left(1 + \frac{p_i N_a |J(f_i)|^2 [e^{(\hbar f_i/KT)} - 1]}{B\hbar f_i}\right) \tag{26}$$

in which, $(p_i N_a)/B$ is the power density of the ith sub-band, $|J(f_i)|^2$ is the equivalent channel gain at the frequency f_i, $\hbar = 6.625 \times 10^{-34} Joule.sec$ is the Planck's constant, $K = 1.3806 \times 10^{-23} Joule/Kelvin$ is the Boltzmann's constant, and the $T = 290 Kelvin$ denotes system temperature.

In Fig. 2, the achievable rate results of different methods are plotted by increasing time. As observed, the proposed DCBS achieves better rate performance. For instance, the achievable rate can be achieved by the proposed DCBS with $\delta = 0$ and $v = 15$ km/h is approximately 350 Gbps higher than that of the InFous scheme. Furthermore, the achievable rates are also obtained at different vehicle velocity and angle as shown in Fig. 3 and Fig. 4, respectively. Fig. 3 shows that at time $t = 2$ s and $\delta = 0$, the proposed DCBS can provide achievable rate about 421 Gbps over the InFous method. In Fig. 4, compared to the InFous method, the proposed DCBS can increase the achievable rate by about 382 Gbps at time $t = 2$ s and $v = 35$ km/h. This is expected since the InFocus method using the NLFM designed on the stationary phase principle has the poor Doppler tolerance. The proposed DCBS design the chirp

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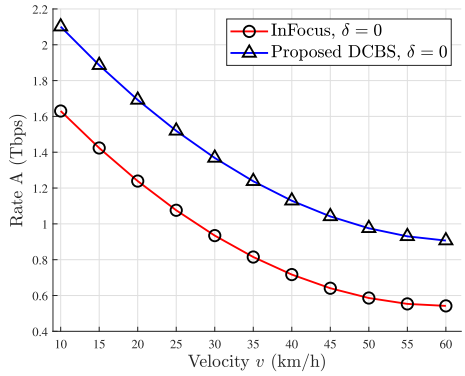

Fig. 3. Achievable rate at different vehicle velocity v at time $t = 2$ s along the boresight direction.

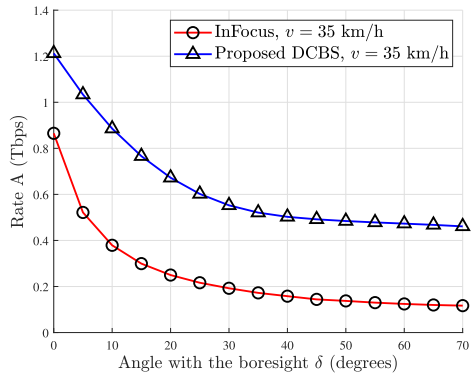

Fig. 4. Achievable rate at different angle δ at time $t = 2$ s and $v = 35$ km/h.

signal using the hybrid-FM approach to control the Doppler sensitivity, which is suited for high-mobility scenarios.

V. Conclusion

In this letter, we have proposed a closed-form beamforming scheme DCBS to realize high data transmission rate in the near-field for mmWave Internet-of-vehicle. Our proposed DCBS, which is demonstrated by the numerical results, has the ability to achieve high data rate at different angle and speed of vehicle. Hence, it works with high efficiency and remarkable achievements to mitigate the misfocus effect and enhance the Doppler tolerance in the near-field regime, which is suitable for implementing in the high-mobility communication scenarios. In the future work, the reconfigurable intelligent surface can be introduced to assist to joint active and passive beamforming strategies, which controls the radio propagation dynamically with low energy consumption in the near-field region for mmWave Internet-of-vehicle communication.

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